



ECEN 314 - Signals and Systems

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Module 4 - Systems in the Time Domain

Learning Objectives



- Know the 6 basic properties of systems and be able to tell if a given system possesses those properties.
- Understand what the convolutional integral is and how it relates the input and output of a continuous time LTI system.
- Understand what the convolutional sum is and how it relates the input and output of a discrete time LTI system.



In general, a system is a device that takes an input (signal) and produces an output (signal). The following are properties that can be used to characterize/classify systems.

Property #1: Memory – A system has memory if $y(t_o)$ depends on values of $x(t)$ only for $t \neq t_o$.

- ✓ If $y(t_o)$ depends only on $x(t_o)$ then the system is memoryless.
- ✓ If we need to know values of $x(t)$ at times other than $t = t_o$ in order to form $y(t_o)$, then the system has memory.

Property #2: Causality – A system is causal if $y(t_o)$ depends on values of $x(t)$ only for $t \leq t_o$.

- ✓ Current output depends on current and past inputs, not future inputs.
- ✓ All buildable systems must be causal (unless you own a time machine).



Property #3: Invertibility – A system is invertible if each distinct input $x(t)$ produces a distinct output $y(t)$.

- ✓ If you observe the output, can you always determine the input?
- ✓ For an invertible system, two different inputs do not produce the same output.

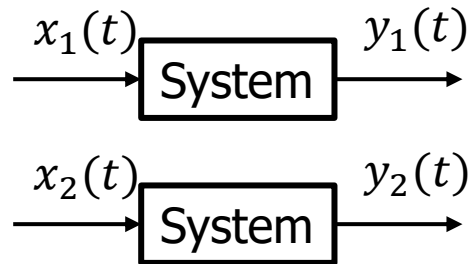
Property #4: Stability – A system is stable if for any $x(t)$ such that $|x(t)| < c_1$, then $|y(t)| < c_2$, for any finite constants c_1 and c_2 .

- ✓ bounded inputs produce bounded outputs.

If these properties don't hold, it is generally easy to show that. All you need is one counterexample.

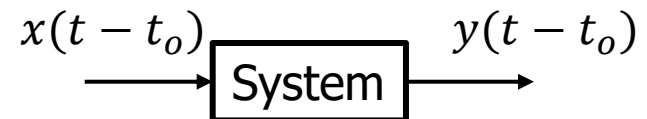
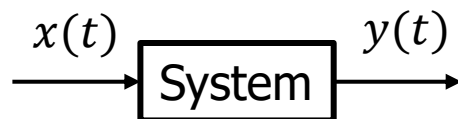
Proving that these properties do hold can be more difficult. You have to show it is true for every possible input/output.

Systems



Property #5: Linearity – If $x_1(t) \rightarrow y_1(t)$ and $x_2(t) \rightarrow y_2(t)$, the system is linear if $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$.

✓ In other words, this means that for a linear system, superposition holds.



Property #6: Time-Invariance – A system is time invariant if for any input $x(t)$ that produces an output $y(t)$ then the input $x(t - t_o)$ produces $y(t - t_o)$.

✓ Time shifted inputs produce time shifted outputs.

Example 4.1: Suppose a system has an input/output relationship given by

$$y(t) = \int_{-\infty}^t x(u) du.$$

This is known as an integrator. Determine if this system possesses any of the six properties listed on the previous slide.

Example 4.2: Suppose a system has an input/output relationship given by

$$y(t) = x^2(t).$$

Determine if this system possesses any of the six properties of systems.

Example 4.3: Suppose a system has an input/output relationship given by the differential equation

$$\frac{dy}{dt} + y(t) = x(t).$$

Determine if this system possesses any of the six properties of systems.

Example 4.4: Suppose a discrete-time system has an input/output relationship given by the differential equation

$$y[n] = x[n] \cdot \cos(n\pi/4).$$

Determine if this system possesses any of the six properties of systems.

An Important Class of Systems



Suppose a system has an input/output relationship described by a generic ordinary linear constant coefficient differential equation

$$\sum_{m=0}^M b_m \frac{d^m y}{dt^m} = \sum_{n=0}^N a_n \frac{d^n x}{dt^n}.$$

The output of the system can be broken into two parts:

- Natural response – Due to initial conditions (found assuming $x(t) = 0$),
- Forced Response – Due to the input (found assuming all ICs are 0).

For most of this course, we will concern ourselves with the forced response of such systems and will assume that the initial conditions are all zero.

It can be shown that such a system is linear and time-invariant (LTI). The other 4 properties (memory, invertibility, causality, stability) may or may not hold depending on the coefficients of the ODE.

For a general LTI system, it is possible to determine the output of the system in terms of the input. The derivation follows.

Output of LTI Systems



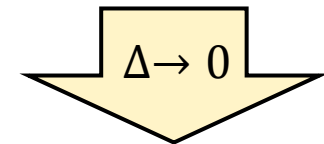
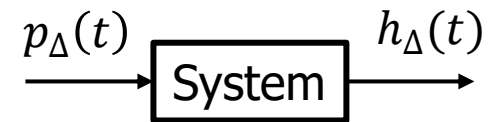
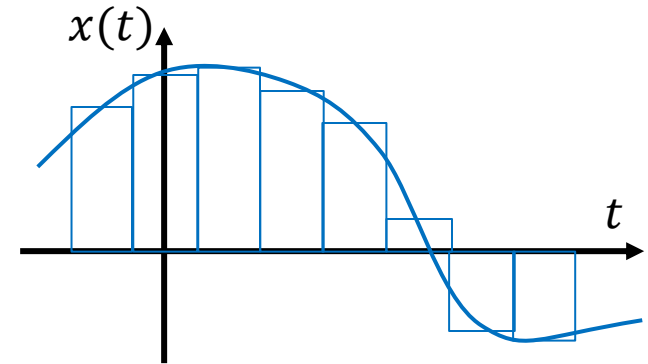
To start the derivation, we approximate the input with a series of rectangular pulses. Let $p_{\Delta}(t) = \frac{1}{\Delta} \text{rect}\left(\frac{t}{\Delta}\right) \rightarrow$ rectangle of height $\frac{1}{\Delta}$, width Δ and area 1. Then

$$x(t) \cong \sum_{k=-\infty}^{\infty} x(k\Delta) \cdot p_{\Delta}(t - k\Delta) \cdot \Delta.$$

Note that as $\Delta \rightarrow 0$,

- ✓ The approximation becomes exact, and
- ✓ $p_{\Delta}(t) \rightarrow \delta(t)$.

Let $h_{\Delta}(t)$ be the response (output) of the system to an input of $p_{\Delta}(t)$. Also, let $h(t) = \lim_{\Delta \rightarrow 0} h_{\Delta}(t)$ so that $h(t)$ is the response of the system to a unit impulse at the input. We call $h(t)$ the *impulse response* of the system.



Output of LTI Systems



Starting with $p_{\Delta}(t) \rightarrow h_{\Delta}(t)$, then using the properties of linearity and time invariance:

$$p_{\Delta}(t - k\Delta) \rightarrow h_{\Delta}(t - k\Delta) \text{ (due to time invariance)}$$

$$\Delta \cdot x(k\Delta) \cdot p_{\Delta}(t - k\Delta) \rightarrow \Delta \cdot x(k\Delta) \cdot h_{\Delta}(t - k\Delta) \text{ (due to linearity)}$$

$$\sum_{k=-\infty}^{\infty} \Delta \cdot x(k\Delta) \cdot p_{\Delta}(t - k\Delta) \rightarrow \sum_{k=-\infty}^{\infty} \Delta \cdot x(k\Delta) \cdot h_{\Delta}(t - k\Delta) \text{ (due to linearity)}$$

Passing to the limit as $\Delta \rightarrow 0$, the input becomes $x(t)$ and the output becomes

$$y(t) = \int_{-\infty}^{\infty} x(v)h(t - v)dv.$$

This is known as a *convolution* integral and we often use the shorthand notation

$$y(t) = x(t) * h(t).$$

This result tells us that if we can determine the impulse response of a system, we can then determine the response of the system to any arbitrary input.

Convolution in Discrete Time



A similar result can be developed for discrete time systems.

- Let $h[n]$ be the discrete time impulse response of the system.
- $\rightarrow h[n]$ is the output when the input is $\delta[n]$.

✓ Then, for a general input, $x[n]$, the output can be written as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

✓ For discrete time systems, convolution is a sum rather than an integral.

✓ Again, we often use the shorthand notation

$$y[n] = x[n] * h[n].$$

The next set of notes will elaborate on how to evaluate convolution integrals/sums.